Weak-Star Basic Sequences in $C_p(K)'$: Verification, Literature Status, and an Exact Reduction

Research report prepared from the supplied partial proof

June 22, 2026

Abstract

The supplied argument is correct: if a compact Hausdorff space K contains a nontrivial convergent sequence, then the finitely supported measures $\delta_{x_n} - \delta_x$ form a weak-star basic sequence in $C_p(K)'$. A literature search did not locate a proof or counterexample for the full problem posed as Problem 12 in Kąkol–López-Pellicer–Śliwa (2026).

The main additional result of this report is an exact reduction. Combining the known quotient characterization of the finitely supported Josefson–Nissenzweig property with an elementary norm-rigidity observation, we prove that, for every infinite compact Hausdorff space K, the dual $C_p(K)'$ contains a weak-star basic sequence if and only if $C_p(K)$ has an infinite-dimensional metrizable quotient. Thus the 2026 weak-star basic-sequence question is equivalent to the older metrizable-quotient problem for $C_p(K)$. The same equivalence in fact holds for every infinite Tychonoff space. In the compact setting it yields a complete positive answer for every non-Efimov compactum and for every compactum with the finitely supported Josefson–Nissenzweig property, but it does not settle the remaining Efimov case. The reduction is proved here and was not located in the searched literature; it should therefore be regarded as an unreviewed deduction.

1 Problem, notation, and conclusions

For a Tychonoff space K, $C_p(K)$ denotes $C(K)$ with the topology of pointwise convergence. Its continuous dual is

$$L_p(K) = C_p(K)' = \Delta(K) = \left\{ \sum_{j=1}^{m} a_j \delta_{x_j} : m < \infty,\ a_j \in \mathbb{R},\ x_j \in K \right\},$$

the vector space of finitely supported signed measures. It is equipped with

$$\sigma(L_p(K), C_p(K)) = \sigma(\Delta(K), C(K)),$$

which we call the weak-star topology.

A sequence is weak-star basic if it is a Schauder basis of its weak-star closed linear span. In particular, its coefficient functionals must be continuous for the subspace weak-star topology.

The source paper asks:

> Does $C_p(K)'$ admit a weak-star basic sequence for every infinite compact space K?

The conclusions reached here are as follows.

(i) The supplied convergent-sequence proof is valid without substantive correction; see section 2.
(ii) No later proof or counterexample was found in the literature search described in section 4. The arXiv record of the source still has only its January 2026 first version.
(iii) The problem is positive for every non-Efimov compactum; equivalently, it is positive whenever K contains a nontrivial convergent sequence or a copy of $\beta\mathbb{N}$.
(iv) Most importantly, the problem is equivalent to the existence of an infinite-dimensional metrizable quotient of $C_p(K)$; see theorem 6.1.
(v) Both global problems reduce to infinite separable compacta: any infinite compactum contains an infinite separable closed subcompactum, and restriction transfers a positive answer upward.

2 Verification of the supplied theorem

Theorem 2.1 (Convergent-sequence case)**.** *Let K be a compact Hausdorff space. Suppose that (x_n) is a sequence of pairwise distinct points converging to $x \in K \setminus \{x_n : n \in \mathbb{N}\}$. Then*

$$\mu_n = \delta_{x_n} - \delta_x \qquad (n \in \mathbb{N})$$

is a weak-star basic sequence in $L_p(K)$.

Proof. Put

$$A = \{x\} \cup \{x_n : n \in \mathbb{N}\}, \qquad M = \operatorname{span}\{\mu_n : n \in \mathbb{N}\}.$$

The set A is compact and hence closed in K.

We first prove that M is weak-star closed in the *relative* dual $L_p(K)$. Let

$$\nu = \sum_{j=1}^{m} a_j \delta_{t_j} \in \overline{M}^{w^*},$$

where the points t_j are distinct and all a_j are nonzero. If $t_j \notin A$, normality of K gives $f \in C(K)$ such that

$$f(t_j) = 1, \qquad f|_{A\cup(\{t_1,\dots,t_m\}\setminus\{t_j\})} = 0.$$

Every member of M annihilates f, whereas $\nu(f) = a_j \neq 0$, a contradiction. Hence $\operatorname{supp}\nu \subseteq A$.

Since ν has finite support, for some N it has the form

$$\nu = a\delta_x + \sum_{n=1}^{N} b_n \delta_{x_n}.$$

Every member of M annihilates the constant function 1, and therefore so does ν. Thus $a + \sum_{n=1}^{N} b_n = 0$, whence

$$\nu = \sum_{n=1}^{N} b_n(\delta_{x_n} - \delta_x) = \sum_{n=1}^{N} b_n \mu_n \in M.$$

This proves that M is weak-star closed.

The vectors μ_n are linearly independent, so every member of M has a unique finite expansion. For fixed n, the point x_n is isolated in A. Consequently, by normality there exists $f_n \in C(K)$ such that

$$f_n(x_n) = 1, \qquad f_n|_{A\setminus\{x_n\}} = 0.$$

For every finite sum $\sum_k c_k\mu_k$ we then have

$$\left(\sum_k c_k\mu_k\right)(f_n) = c_n.$$

Hence the nth coefficient functional is the restriction to M of the weak-star continuous functional $\nu \mapsto \nu(f_n)$. Finally, every basis expansion in M is finite, so its sequence of partial sums is eventually constant. Therefore (μ_n) is a Schauder basis of the weak-star closed space M. □

Remark 2.2 (What was checked)**.** *The key point is that closure is taken inside $L_p(K)$, whose elements are already finitely supported. The proof does not claim that M is weak-star closed in the full Banach dual $C(K)^*$, where limits may be arbitrary Radon measures. With this relative-closure convention, every step of the supplied argument is correct.*

The following elementary formula gives another view of the closed-span argument.

Lemma 2.3 (Closure of point masses)**.** *Let X be compact Hausdorff and let $D \subseteq X$. Inside $L_p(X)$,*

$$\overline{\operatorname{span}\{\delta_d : d \in D\}}^{w^*} = \operatorname{span}\{\delta_x : x \in \operatorname{cl}_X D\}.$$

Proof. The annihilator of $\operatorname{span}\{\delta_d : d \in D\}$ in $C(X)$ is the set of continuous functions vanishing on D, hence the set of functions vanishing on $\operatorname{cl}_X D$. Taking the annihilator once more gives all finitely supported measures supported on $\operatorname{cl}_X D$. Equivalently, one may separate each point outside $\operatorname{cl}_X D$ from $\operatorname{cl}_X D$ by a continuous function. □

In theorem 2.1, M is exactly the space of finite measures supported on A and having total mass zero. Both conditions are weak-star closed.

3 Transfer and quotient principles

Lemma 3.1 (Restriction to a closed compact subspace)**.** *Let X be compact Hausdorff and $K \subseteq X$ closed. The restriction map*

$$R : C_p(X) \longrightarrow C_p(K), \qquad Rf = f|_K,$$

is a continuous open surjection. Its adjoint

$$R' : L_p(K) \longrightarrow L_p(X)$$

is a weak-star topological embedding with weak-star closed range. Therefore, if $L_p(K)$ has a weak-star basic sequence, then so does $L_p(X)$.

Proof. Surjectivity follows from the Tietze extension theorem. To see openness, let $F \subseteq X$ be finite and let

$$U = \{f \in C(X) : |f(t)| < \varepsilon_t \text{ for every } t \in F\}$$

be a basic zero-neighborhood. Then $R(U)$ contains the basic neighborhood of zero in $C_p(K)$ imposing the same inequalities on $F \cap K$. Indeed, if $g \in C(K)$ satisfies these inequalities, define a continuous function on $K \cup (F \setminus K)$ by taking the value g on K and 0 on $F \setminus K$, and extend it to X.

The assertions about the adjoint are the standard dual consequences of an open quotient map. Concretely, its range consists of the finitely supported measures supported on K, which is weak-star closed by point separation. □

Corollary 3.2 (Reduction to separable compacta)**.** *Each of the following global assertions is equivalent to its restriction to infinite separable compact Hausdorff spaces:*

(a) $L_p(K)$ *contains a weak-star basic sequence;*
(b) $C_p(K)$ *has an infinite-dimensional metrizable quotient.*

In particular, if a counterexample to either assertion exists, then a separable compact counterexample exists.

Proof. Only the upward implication requires proof. Given an infinite compact X, choose a countably infinite set $D \subseteq X$ and put $K = \mathrm{cl}_X D$. Then K is infinite, compact, closed in X, and separable. For (a), apply theorem 3.1. For (b), compose the open restriction map $C_p(X) \to C_p(K)$ with an open quotient of $C_p(K)$. $\square$

Lemma 3.3 (Quotients of weak spaces)**.** *Let* $(E, \sigma(E, F))$ *be a separated weak locally convex space and let* $N \subseteq E$ *be closed. The quotient topology on* E/N *is*

$$\sigma(E/N, N^{\perp}), \qquad N^{\perp} = \{\varphi \in F : \varphi|_N = 0\}.$$

In particular, E/N *is metrizable if and only if* $N^{\perp}$ *has countable Hamel dimension.*

Proof. Every member of $N^{\perp}$ descends to a quotient-continuous functional, so the quotient topology is at least as strong as the displayed weak topology. Conversely, take a weak neighborhood in E determined by $\varphi_1, \ldots, \varphi_m \in F$. Let $A : E \to \mathbb{R}^m$ be $A(x) = (\varphi_1(x), \ldots, \varphi_m(x))$. The saturation of that neighborhood by N is the inverse image under A of an open set saturated by the finite-dimensional subspace $A(N)$. Passing to $\mathbb{R}^m/A(N)$ shows that this saturation contains a neighborhood determined by finitely many linear combinations of the φ_j which vanish on N. Thus the quotient topology has a base of $\sigma(E/N, N^{\perp})$-neighborhoods. If $N^{\perp}$ has a countable Hamel basis, that basis gives a countable family of seminorms generating the quotient topology. Conversely, suppose that the weak quotient is metrizable. Choose a countable zero-neighborhood base and, inside each member, a basic weak neighborhood determined by finitely many elements of $N^{\perp}$. The union G of those finite sets is countable. The family G therefore generates the quotient topology. For any $\psi \in N^{\perp}$, continuity at zero supplies a finite set $H \subseteq G$ and a constant $C > 0$ such that

$$|\psi(x + N)| \le C \max_{\varphi \in H} |\varphi(x)| \qquad (x \in E).$$

Thus ψ vanishes on $\bigcap_{\varphi \in H} \ker \varphi$ and hence, by finite-dimensional linear algebra, belongs to $\operatorname{span} H$. Consequently $N^{\perp} = \operatorname{span} G$. $\square$

4 Literature status and the known positive region

The exact source is Kąkol–López-Pellicer–Śliwa, *On weak*-basic sequences in duals and biduals of spaces* $C(X)$ *and Quojections* [1]. Its Problem 12 is the question studied here, and its Theorem 17 proves that an infinite-dimensional metrizable quotient of $C_p(X)$ yields a weak-star basic sequence in $C_p(X)'$. The arXiv record still lists only the version submitted on January 27, 2026. A June 2, 2026 seminar announcement by Kąkol states both that the two-disjoint-copies property yields such a quotient and that the metrizable-quotient question for arbitrary infinite compacta remains a long-standing open problem [7].

The older paper of Banakh–Kąkol–Śliwa proves that if a compact K is not Efimov, then $C_p(K)$ has a quotient isomorphic, with the product topology, to c_0 or ℓ_∞ [2, Corollary 3]. Since both $(c_0)_p$ and $(\ell_\infty)_p$ have their canonical Schauder bases in the product topology, Theorem 17 of [1] immediately gives the following.

Corollary 4.1 (Non-Efimov compacta)**.** *If an infinite compact Hausdorff space K contains either a nontrivial convergent sequence or a copy of $\beta\mathbb{N}$, then $L_p(K)$ has a weak-star basic sequence.*

The first alternative is also covered constructively by theorem 2.1. In the second alternative, one may use finite block averages on the C^*-embedded copy of $\mathbb{N}$; they are the adjoints of the coordinate functionals of an open quotient onto $(\ell_\infty)_p$.

A second large positive class comes from the finitely supported Josefson–Nissenzweig property.

Definition 4.2. *A compact space K has the* finitely supported Josefson–Nissenzweig property *(fsJNP) if there is a sequence $(\nu_n) \subseteq \Delta(K)$ such that*

$$\|\nu_n\| = 1 \quad (n \in \mathbb{N}), \qquad \nu_n(f) \longrightarrow 0 \quad (f \in C(K)).$$

Banakh–Kąkol–Śliwa proved that the fsJNP is equivalent to the existence of a quotient of $C_p(K)$ isomorphic to $(c_0)_p$, and also to the existence of a complemented copy of $(c_0)_p$ in $C_p(K)$ [3, Theorem 1]. Consequently:

Corollary 4.3. *Every compact space with the fsJNP has a weak-star basic sequence in its C_p-dual.*

Kąkol–Sobota–Zdomskyy proved that many classical consistent Efimov spaces, including inverse limits built from simple extensions, have the fsJNP [4, Corollary 6.12]. They also note that consistently there are Efimov compacta with the Grothendieck property and hence without the fsJNP [4, p. 16]. Thus neither the Efimov property nor the failure of the fsJNP is by itself a known obstruction. Indeed, the June 2026 seminar announcement reports that the Brech and Sobota–Zdomskyy compacta of Efimov type have the two-disjoint-copies property and therefore have metrizable C_p-quotients [7].

The search covered the exact problem wording, the exact paper title and its arXiv record, later arXiv papers using the terms "weak-star basic sequence", "$C_p(X)$", "metrizable quotient", and "Efimov", and the citation chains of the principal 2018–2026 papers. No full proof or counterexample was found through June 22, 2026. This is necessarily a search report rather than a proof of nonexistence of an unpublished result.

5 Sequential norm rigidity without the fsJNP

The absence of the fsJNP has a useful elementary consequence. It is also the sequential core of the equivalence between the lack of the fsJNP and the Δ- and ℓ_1-Grothendieck properties established in [4, Theorem 4.1].

Lemma 5.1 (Sequential norm rigidity)**.** *Assume that a compact space K does not have the fsJNP. If $(\nu_n) \subseteq \Delta(K)$ and $\nu \in \Delta(K)$ satisfy*

$$\nu_n \longrightarrow \nu \quad \text{in } \sigma(\Delta(K), C(K)),$$

then $\|\nu_n - \nu\| \to 0$ in total variation norm.

Proof. Otherwise, after passing to a subsequence, there is an $\varepsilon > 0$ such that $\|\nu_n - \nu\| \geq \varepsilon$ for every n. Then

$$\lambda_n = \frac{\nu_n - \nu}{\|\nu_n - \nu\|}$$

is a norm-one sequence in $\Delta(K)$ and $\lambda_n(f) \to 0$ for every $f \in C(K)$, which is precisely an fsJN-sequence. This contradicts the assumption. □

The next proposition is the main technical observation of this report.

Proposition 5.2 (Basic expansions become norm convergent)**.** *Let K have no fsJNP, and let (μ_n) be a weak-star basic sequence in $L_p(K)$. Let*

$$M = \overline{\operatorname{span}\{\mu_n : n \in \mathbb{N}\}}^{w^*} \subseteq L_p(K), \qquad S = \bigcup_{n\in\mathbb{N}} \operatorname{supp}\mu_n.$$

Then S is countable and every $\mu \in M$ is supported on S. More precisely, if

$$\mu = \sum_{n=1}^{\infty} a_n \mu_n$$

is its basis expansion, then the partial sums converge to μ in total variation norm. Consequently,

$$M \subseteq \Delta(S),$$

so M has countable Hamel dimension.

Proof. The set S is a countable union of finite sets. Fix $\mu \in M$ and let

$$s_N = \sum_{n=1}^{N} a_n \mu_n.$$

By the basis property, $s_N \to \mu$ in $\sigma(\Delta(K), C(K))$. Hence theorem 5.1 gives $\|s_N - \mu\| \to 0$.

Every s_N is supported on S. Since S is countable, it is Borel, and

$$\big\|\mu|_{K\setminus S}\big\| \le \|\mu - s_N\| \longrightarrow 0.$$

Thus μ is supported on S. Since $\mu \in L_p(K) = \Delta(K)$ is finitely supported, $\mu \in \Delta(S)$. Because S is countable, $\Delta(S)$ has a countable Hamel basis. □

Corollary 5.3 (Metrizable quotient in the no-fsJNP case)**.** *If K has no fsJNP and $L_p(K)$ has a weak-star basic sequence, then $C_p(K)$ has an infinite-dimensional metrizable quotient.*

Proof. Let M be the weak-star closed span of the basic sequence and put

$$N = M_\perp = \{f \in C(K) : \mu(f) = 0 \text{ for all } \mu \in M\}.$$

By the bipolar theorem for the dual pair $(C(K), L_p(K))$ and the weak-star closedness of M,

$$N^\perp = M.$$

By theorem 5.2, M has countable Hamel dimension. Applying theorem 3.3 to $C_p(K) = (C(K), \sigma(C(K), L_p(K)))$, the quotient $C_p(K)/N$ is metrizable. It is infinite-dimensional because its continuous dual M is infinite-dimensional. □

6 Exact equivalence with the metrizable-quotient problem

Theorem 6.1 (Main reduction)**.** *For every infinite compact Hausdorff space K, the following are equivalent.*

(1) *$L_p(K) = C_p(K)'$ contains a weak-star basic sequence.*
(2) *$C_p(K)$ has an infinite-dimensional metrizable quotient.*

(3) $C_p(K)$ *has an infinite-dimensional metrizable quotient possessing a Schauder basis.*
(4) $L_p(K)$ *contains an infinite-dimensional weak-star closed linear subspace of countable Hamel dimension.*

Proof. First, (2) and (4) are equivalent. If $q : C_p(K) \to E$ is an infinite-dimensional metrizable quotient and $N = \ker q$, then theorem 3.3 identifies E with the weak space $\sigma(C(K)/N, N^{\perp})$ and shows that $N^{\perp}$ has countable Hamel dimension. It is infinite-dimensional and weak-star closed in $L_p(K)$.

Conversely, let $M \subseteq L_p(K)$ be infinite-dimensional, weak-star closed, and of countable Hamel dimension. Put $N = M_{\perp}$. Then $N^{\perp} = M$, and theorem 3.3 shows that $C_p(K)/N$ is an infinite-dimensional metrizable quotient.

The implication (2)⇒(3) is Theorem 17 of [1]: every infinite-dimensional metrizable quotient of $C_p(K)$ has a Schauder basis. For (3)⇒(1), let (e_n) be such a basis in the quotient E, with coordinate functionals (e_n^*). If P_N denotes the Nth basis projection, then for every $\varphi \in E'$,

$$\sum_{n=1}^{N} \varphi(e_n)e_n^* = \varphi \circ P_N \longrightarrow \varphi \quad \text{in } \sigma(E', E).$$

The coefficients are uniquely recovered by evaluation at the e_n, so (e_n^*) is a weak-star Schauder basis of E'. The adjoint of the open quotient map embeds E' as a weak-star closed subspace of $L_p(K)$.

Assume (1). If K has the fsJNP, the theorem of Banakh–Kąkol–Śliwa gives a quotient of $C_p(K)$ isomorphic to $(c_0)_p$ [3, Theorem 1], so (2) holds. If K does not have the fsJNP, theorem 5.3 again gives (2). □

Corollary 6.2 (Equivalence of the two global problems)**.** *The following assertions are equivalent.*

(a) *For every infinite compact Hausdorff* K*,* $C_p(K)'$ *has a weak-star basic sequence.*
(b) *For every infinite compact Hausdorff* K*,* $C_p(K)$ *has an infinite-dimensional metrizable quotient.*

Thus Problem 12 of the 2026 paper is not merely related to the earlier metrizable-quotient problem: it is exactly equivalent to it.

Remark 6.3 (Tychonoff extension)**.** *Compactness is not used in the argument of theorems 5.1 to 5.3. For an arbitrary Tychonoff space* X*, the dual of* $C_p(X)$ *is still* $\Delta(X)$*, with* $\|\sum a_j\delta_{x_j}\| = \sum|a_j|$*. Moreover, the JNP–*$(c_0)_p$ *quotient characterization [3] and Theorem 17 of [1] both hold for Tychonoff spaces. Hence the four conditions in theorem 6.1 are in fact equivalent for every infinite Tychonoff space* X*.*

7 What a counterexample would have to look like

Combining theorem 6.1 with the known positive results gives a sharp necessary profile.

Corollary 7.1. *If* K *is a counterexample to Problem 12, then:*

(i) K *is an Efimov compactum: it contains neither a nontrivial convergent sequence nor a copy of* $\beta\mathbb{N}$*;*
(ii) K *has the* ℓ_1*-Grothendieck property, hence no fsJNP;*
(iii) $C_p(K)$ *has no infinite-dimensional metrizable quotient;*
(iv) *in particular,* $C_p(K)$ *has no quotient isomorphic to* $(c_0)_p$ *or* $(\ell_\infty)_p$*;*

(v) *K fails the two-disjoint-copies criterion of Banakh–Kąkol–Śliwa.*

Conversely, any infinite compact K satisfying item (iii) *is a counterexample.*

The last converse is simply theorem 6.1. Moreover, theorem 3.2 shows that, if any counterexample exists, one can choose a separable one; in fact, the closure of every countably infinite subset of a counterexample is again a counterexample. At present, no such compactum is known from the searched literature.

8 Why several natural approaches stop short

A closed copy of $(c_0)_p$ is not enough

Recent work shows that every infinite compact K supplies natural closed copies of $(c_0)_p$ in $C_p(K)$, obtained from pairwise disjoint open sets and bump functions [5]. But inclusion of a closed subspace $F \hookrightarrow C_p(K)$ gives a *surjection* $C_p(K)' \to F'$ by restriction, not a canonical weak-star closed embedding of F' into $C_p(K)'$. Such an embedding follows from a continuous projection onto F, and complementability is precisely the extra ingredient supplied by the fsJNP. Therefore the closed-subspace theorem alone does not settle the dual problem.

An infinite metrizable image points in the wrong direction

Every infinite compact K has an infinite compact metrizable continuous image: choose countably many continuous functions with infinite-dimensional linear span and take their diagonal map. The pullback embeds the corresponding C_p-space into $C_p(K)$. However, a basic sequence in the dual transfers from a *quotient* or from a closed subspace of the compactum via theorem 3.1; an arbitrary continuous image gives the opposite variance. A continuous section or averaging operator would repair this, but such an operator is not available for a general compact surjection.

The product result does not descend

The complemented-copy theorem for $C_p(X \times Y)$ gives a weak-star basic sequence in $C_p(X \times Y)'$ for infinite X and Y [6]. Restriction to the diagonal is a quotient $C_p(X \times X) \to C_p(X)$, so its adjoint embeds $C_p(X)'$ into the product dual. This transfers a solution from X to $X \times X$, not from the product back to X. A basic sequence found somewhere in the larger dual need not lie in the diagonal-supported range.

9 Final assessment

The supplied theorem is correct and gives a particularly transparent explicit basis. It was already implicit in the metrizable-quotient machinery and in earlier C_p-results, so it should be presented as a direct construction rather than as a solution of the full problem.

No full proof or counterexample was located. The strongest additional conclusion obtained here is theorem 6.1: weak-star basic sequences in $C_p(X)'$ are equivalent to infinite-dimensional metrizable quotients of $C_p(X)$, already for arbitrary Tychonoff X; see theorem 6.3. By theorem 3.2, it is enough to resolve that quotient problem for separable compacta. The remaining task is therefore concentrated entirely in the unresolved separable Efimov case.

References

[1] J. Kąkol, M. López-Pellicer, and W. Śliwa, *On weak*-basic sequences in duals and biduals of spaces* $C(X)$ *and Quojections*, arXiv:2601.19873, 2026.

[2] T. Banakh, J. Kąkol, and W. Śliwa, *Metrizable quotients of* C_p*-spaces*, Topology Appl. **249** (2018), 95–102; arXiv:1804.02552.

[3] T. Banakh, J. Kąkol, and W. Śliwa, *Josefson–Nissenzweig property for* C_p*-spaces*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM **113** (2019), 3015–3030; arXiv:1809.07054.

[4] J. Kąkol, D. Sobota, and L. Zdomskyy, *Grothendieck* $C(K)$*-spaces and the Josefson–Nissenzweig theorem*, Fund. Math. **263** (2023), no. 2, 105–131; arXiv:2207.13990.

[5] J. Kąkol, A. Moltó, and W. Śliwa, *On subspaces of spaces* $C_p(X)$ *isomorphic to spaces* c_0 *and* ℓ_q *with the topology induced from* $\mathbb{R}^{\mathbb{N}}$, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat. **117** (2023), article 154.

[6] J. Kąkol, W. Marciszewski, D. Sobota, and L. Zdomskyy, *On complemented copies of the space* c_0 *in spaces* $C_p(X \times Y)$, Israel J. Math. **250** (2022), 139–177.

[7] J. Kąkol, *The two-disjoint-copies property for compact spaces, homogeneity, and relations with the theory of spaces* C_p *of continuous functions, Part II*, seminar announcement, Adam Mickiewicz University, June 2, 2026 (online announcement).